

Foundation (Jalapeno)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.

- (A) $x = -2, x = -3$
 (B) $x = -1, x = -6$
 (C) $x = 2, x = 3$
 (D) $x = 1, x = 6$
 (E) $x = -2, x = 3$

Q2. Verify the solutions to the equation $x^2 - 7x + 12 = 0$ are $x = 3$ and $x = 4$.

- (A) True
 (B) False
 (C) Maybe
 (D) Not enough information
 (E) None of the above

Q3. A company's profit P is given by the quadratic equation $P = -x^2 + 10x - 20$. Find the value of x when $P = 0$ by factoring.

- (A) $x = 2, x = 10$
 (B) $x = 5, x = 5$
 (C) $x = 10, x = 2$
 (D) $x = 1, x = 9$
 (E) $x = 0, x = 10$

Q4. The quadratic equation $x^2 - 9x + 20 = 0$ has solutions $x = 4$ and $x = 5$. Verify these solutions.

- (A) True
 (B) False
 (C) Maybe
 (D) Not enough information
 (E) None of the above

Q5. The tax T on a person's income I is given by $T = 0.2I^2 - 10I + 50$. Solve for I when $T = 0$.

- (A) $I = 0, I = 25$
 (B) $I = 5, I = 5$
 (C) $I = 25, I = 0$
 (D) $I = 10, I = 10$
 (E) $I = 1, I = 24$

Q6. The stock price S of a company can be modeled by the quadratic equation $S = x^2 - 4x + 5$. Find the value of x when $S = 0$.

- (A) $x = 2\sqrt{5}$
 (B) $x = 2\sqrt{5}$
 (C) $x = 2\sqrt{5}$
 (D) $x = 1, x = 3$
 (E) $x = 0, x = 4$

Q7. An investor's return R is given by $R = -x^2 + 12x - 30$. Solve for x when $R = 0$ by factoring.

- (A) $x = 3, x = 10$
 (B) $x = 5, x = 6$
 (C) $x = 6, x = 5$
 (D) $x = 1, x = 11$
 (E) $x = 0, x = 12$

Q8. The savings S in a bank account are given by $S = x^2 + 2x + 1$. Verify the solution $x = -1$.

- (A) True
 (B) False
 (C) Maybe
 (D) Not enough information
 (E) None of the above

Open Ended Questions (FRQ)**Q9.** Solve the quadratic equation $x^2 - 3x - 4 = 0$ by factoring and verify the solutions. Show all work and explain your answer.**Q10.** A person's portfolio value V can be modeled by $V = -x^2 + 8x + 10$. Find the value of x when $V = 0$. Verify these solutions and explain your answer.

Chef's Tips

- Check that students show all work and explain their answers for free response questions
- Emphasize the importance of verifying solutions to quadratic equations
- Ensure students can factor quadratic equations to solve for the variable